

JEWELRY WORLD

Cloud-Based Java Full Stack E-Commerce Application

A Project Documentation Report

Submitted by

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Technology Stack

Java, JSP, Jakarta Servlets, JDBC, MySQL (Aiven Cloud)

HTML5, CSS3, JavaScript, Bootstrap 5

Apache Tomcat, Maven, Git and GitHub, Render Cloud

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1. Abstract

Jewelry World is a cloud-based full-stack e-commerce application developed using Java Enterprise technologies. The application allows customers to browse jewelry products, create accounts, securely log in, add products to a cart and wishlist, place orders, review products, and manage their profiles.

The administrator module provides complete inventory management, product management, customer order tracking, and administrative control.

The application follows the MVC (Model View Controller) architecture and is deployed on Render using an Aiven Cloud MySQL database.

2. Introduction

Online jewelry shopping has become one of the fastest-growing segments of e-commerce. Jewelry World aims to provide a responsive, secure, and scalable platform that simplifies online jewelry purchases while allowing administrators to efficiently manage inventory and customer orders.

3. Objectives

- Develop a complete online jewelry shopping platform.
- Implement secure user authentication.
- Enable shopping cart functionality.
- Support wishlist management.
- Implement order placement and order tracking.
- Allow customers to submit reviews.
- Build an administrator dashboard.
- Deploy the application on the cloud.

4. System Requirements

Hardware

- Processor: Intel Core i5 or above
- RAM: 8 GB
- Storage: 20 GB

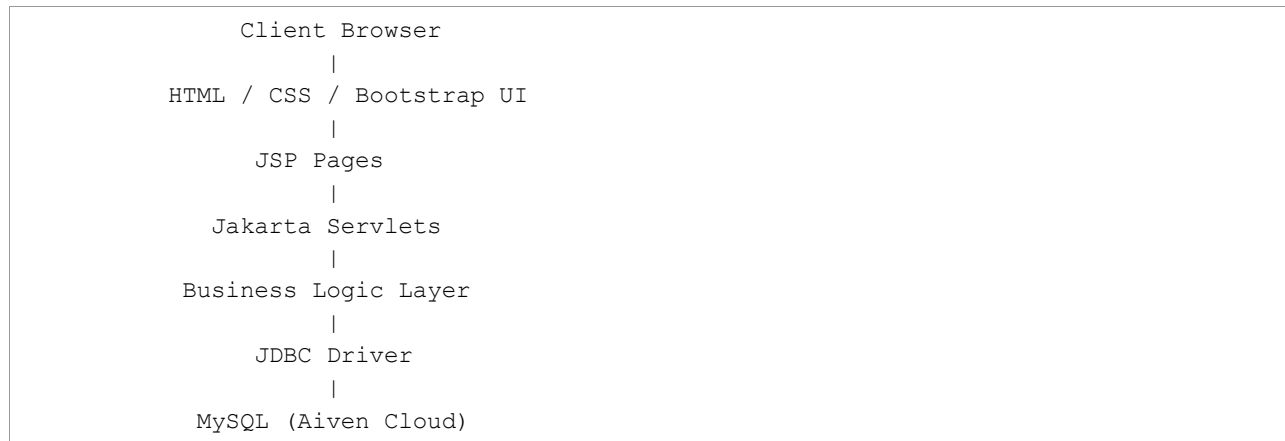
Software

- Java 17
- Apache Tomcat 10

- Maven
- MySQL
- Git
- Eclipse IDE
- Render
- Aiven Cloud Database

5. Software Architecture

The application follows a layered architecture as illustrated below:



6. Functional Modules

Customer Module

- Registration
- Login / Logout
- Product Search
- Category Browsing
- Product Details
- Cart
- Wishlist
- Checkout
- Orders
- Profile
- Product Reviews

Admin Module

- Admin Login

- Dashboard
- Add Products
- Update Products
- Delete Products
- Manage Categories
- Manage Orders
- Manage Customers
- View Reviews

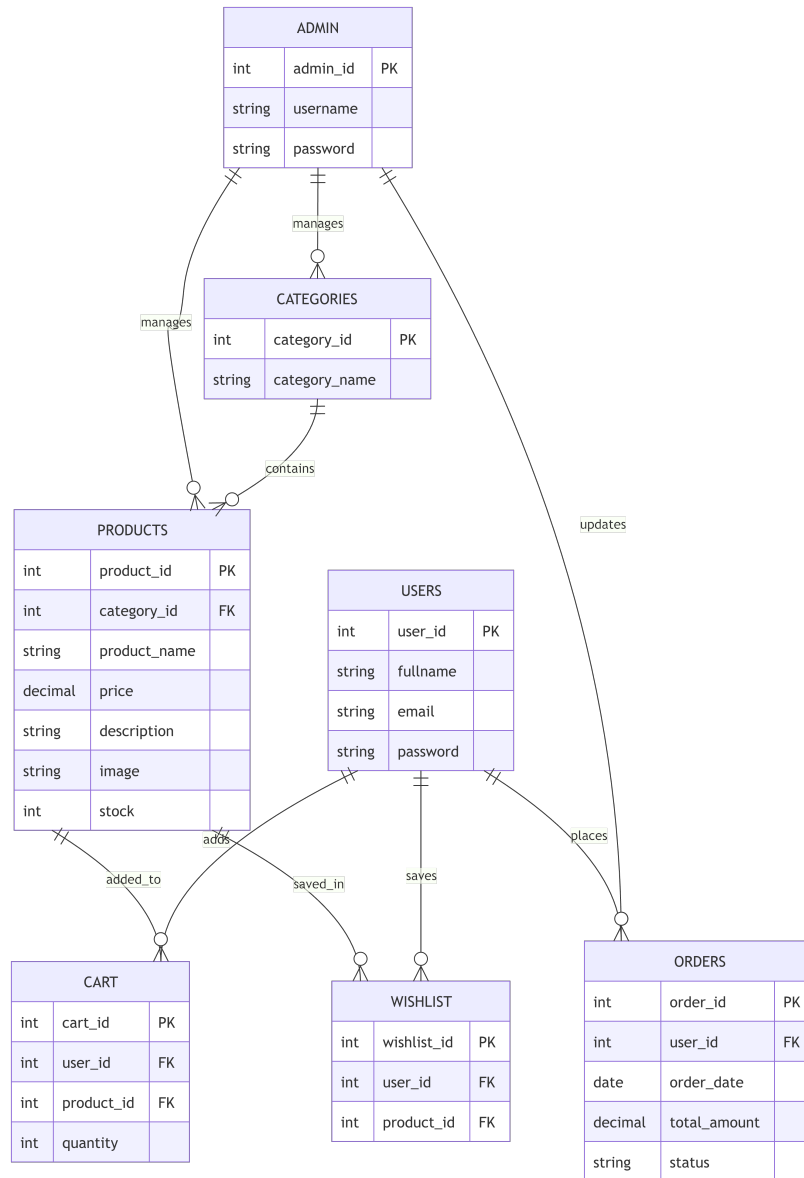
7. Database Design

The application's relational database consists of the following core tables:

Table	Purpose
users	Stores registered customer account details
products	Stores jewelry product catalog information
categories	Stores product category definitions
cart	Tracks items a user has added to their cart
wishlist	Tracks items a user has saved for later
orders	Stores order header information per transaction
order_items	Stores individual line items within an order
reviews	Stores customer product ratings and comments
admin	Stores administrator account credentials

8. ER Diagram

The entity-relationship diagram below shows the core relationships between entities:



9. Database Schema

Users

Column
user_id
fullname
email
password

Column
phone
address

Products

Column
product_id
category_id
product_name
description
price
image
stock

Orders

Column
order_id
user_id
total_amount
status
order_date

Order Items

Column
item_id
order_id
product_id
quantity
price

Wishlist

Column
wishlist_id
user_id
product_id

Cart

Column
cart_id
user_id
product_id
quantity

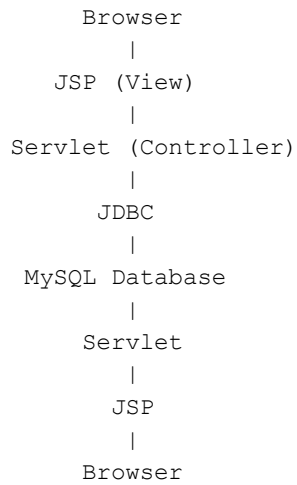
Reviews

Column
review_id
user_id
product_id
rating
comment
review_date

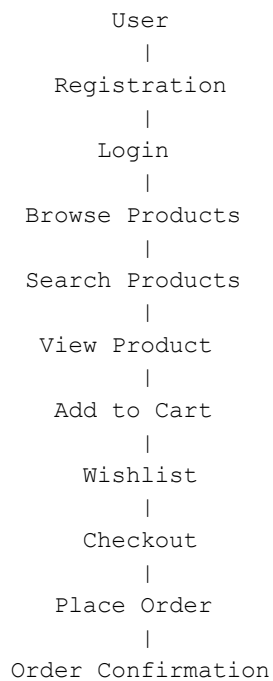
Admin

Column
admin_id
username
password

10. MVC Architecture



11. Application Flow



12. Features

- Secure Login
- Registration
- Session Management
- Product Search

- Category Filtering
- Shopping Cart
- Wishlist
- Product Reviews
- Order History
- Admin Dashboard
- Cloud Database
- Responsive UI
- Cloud Deployment

13. Testing

Functional Testing

- User Registration Pass
- Login Authentication Pass
- Product Search Pass
- Cart Management Pass
- Wishlist Pass
- Checkout Pass
- Orders Pass
- Reviews Pass
- Admin CRUD Pass

Integration Testing

- JDBC Connection Pass
- Database Operations Pass
- Session Management Pass

Deployment Testing

- Render Deployment Pass
- Aiven Database Connectivity Pass

14. Deployment

Layer	Platform
Frontend & Backend	Render
Database	Aiven MySQL Cloud
Version Control	GitHub

15. Future Enhancements

- Payment Gateway Integration (Stripe/PayPal)
- Email Notifications
- JWT Authentication
- Spring Boot Migration
- REST API
- React Frontend
- Product Recommendations using Machine Learning
- AI Chatbot
- Inventory Analytics Dashboard

16. Conclusion

Jewelry World is a cloud-based Java Full Stack e-commerce application developed using JSP, Jakarta Servlets, JDBC, and MySQL. The project demonstrates secure authentication, cloud database integration, session management, product management, shopping cart, wishlist, order processing, and an administrator dashboard following the MVC architecture. The application is deployed on Render with an Aiven Cloud MySQL database, demonstrating practical experience in cloud deployment and enterprise Java web development.

Deployment on Render with Aiven MySQL showcases practical experience in cloud application hosting and database management, making it a strong portfolio project for Java Full Stack Developer and Software Engineer roles.

PROJECT STATISTICS

Programming Language : Java

Architecture : MVC

Database : MySQL (Aiven Cloud)

Deployment : Render Cloud

Version Control : Git & GitHub

Build Tool : Maven

Live Demo: <https://jewelry-world-ecommerce.onrender.com>

Source Code: <https://github.com/swethakanumuri25-cpu/Jewelry-World-ECommerce.git>

APPLICATION SCREENSHOTS

Luxury Jewelry Collection

Discover timeless elegance crafted with love and brilliance.

Shop Collection

Shop By Category



Rings



Necklaces



Earrings



Bracelets

Admin Dashboard

4

Total Products

2

Total Users

3

Total Orders

0

Pending Orders

2

Delivered Orders

\$1697.00

Total Revenue

Manage Products

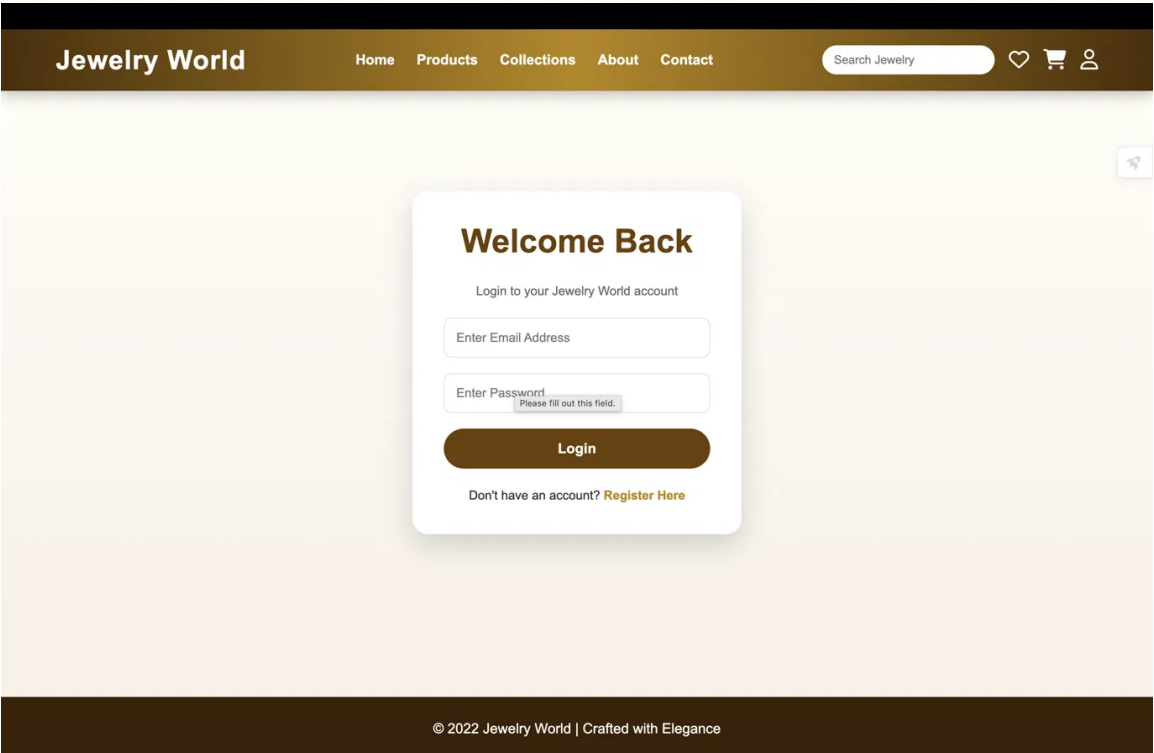
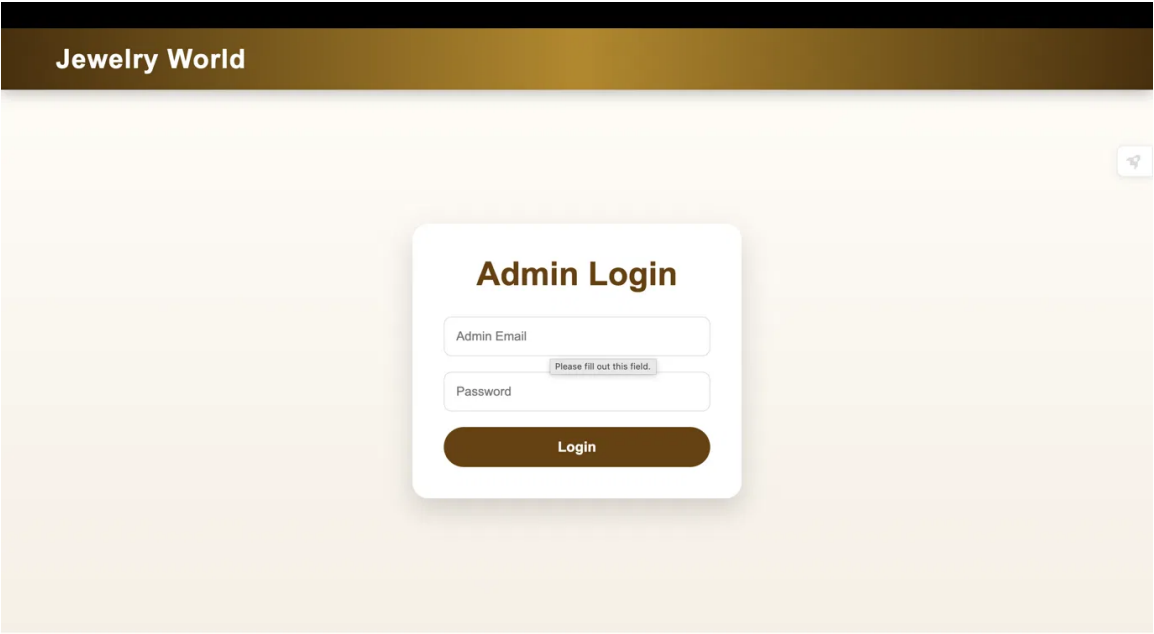
Open

Manage Orders

Open

View Users

Open



Jewelry World

[Home](#)[Products](#)[Collections](#)[About](#)[Contact](#)

Search Jewelry...

Welcome,
Swetha Kanumuri

My Profile

Wishlist

My Orders

Logout

Our Jewelry Collection

All

Rings

Necklaces

Bracelets

Earrings



Diamond Ring

★ 0.0

\$599.0

View Details



Gold Necklace

★ 5.0

\$799.0

View Details



Pearl Earrings

★ 0.0

\$399.0

View Details



Gold Bracelet

★ 0.0

\$499.0

View Details

https://jewelry-world-ecommerce.onrender.com/products.jsp#

Jewelry World Admin

Dashboard

Logout

Manage Products

+ Add Product

ID	Image	Name	Category	Price	Actions
1		Diamond Ring	Ring	\$599.0	Edit Delete
2		Gold Necklace	Necklace	\$799.0	Edit Delete
3		Pearl Earrings	Earrings	\$399.0	Edit Delete
6		Gold Bracelet	Bracelet	\$499.0	Edit Delete

14

Jewelry World Admin

DashboardProductsLogout

Add Product

Product Name

Price

Image Name (example: ring.jpg)

Category

Description

+ Add Product

Jewelry World Admin

DashboardLogout

Manage Orders

ID	Name	Email	Total	Date	Status	Action
3	Swetha Kanumuri	kanumuriswetha9@gmail.com	\$898.0	2026-07-08 16:03:42.0	Delivered	Save
2	Swetha Kanumuri	kanumuriswetha9@gmail.com	\$799.0	2026-07-08 06:35:55.0	Delivered	Save
1	Swetha Kanumuri	kanumuriswetha9@gmail.com	\$599.0	2026-07-07 19:12:07.0	Cancelled	Save

Payment

Select Payment Method

☒ Credit / Debit Card☐ UPI☐ Cash On Delivery[Pay & Place Order](#)**Order Placed Successfully!**

Thank you for shopping with Jewelry World.

[Continue Shopping](#)[View My Orders](#)

Contact Us

We'd love to hear from you.

Get In Touch

**Address**
123 Jewelry Street, Dallas, Texas

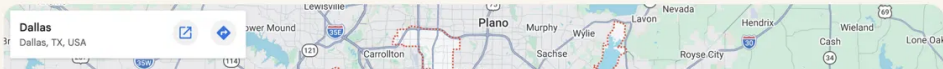
**Phone**
+1 (555) 123-4567

**Email**
support@jewelryworld.com

**Working Hours**
Mon - Sat : 9:00 AM - 7:00 PM

Send Message

Send Message



Registered Users

ID	Full Name	Email
2	Mani	kanumurimani20026@gmail.com
1	Swetha Kanumuri	kanumuriswetha9@gmail.com

Checkout

Please fill out this field.

Place Order

Our Collections

Discover timeless elegance crafted for every occasion.



Rings

Shop Now



Necklaces

Shop Now



Earrings

Shop Now



Bracelets

Shop Now

♥

My Wishlist



Gold Necklace

22K Gold Necklace

\$799.0

Add to Cart

Remove



About Jewelry World

Jewelry World is your trusted destination for elegant, premium-quality jewelry crafted with timeless beauty. Our collections are designed for every special occasion, combining luxury, craftsmanship, and affordability.

Whether you're looking for engagement rings, stylish bracelets, elegant necklaces, or everyday accessories, we ensure every piece reflects sophistication and excellence.



Premium Quality

Finest handcrafted jewelry.



Fast Delivery

Quick and secure shipping.



Certified Products

Authentic and trusted quality.



Customer Satisfaction

Thousands of happy customers.

Jewelry World

HomeProducts

My Profile

Welcome

Swetha Kanumuri

Email : kanumuriswetha9@gmail.com

My Orders

Wishlist

Logout

Jewelry World

HomeProductsProfile

My Orders

Order #3

Name : Swetha Kanumuri

Email : kanumuriswetha9@gmail.com

Address : 903 Avenue C Apt 1302

Total : \$898.0

Ordered On : 2026-07-08 16:03:42.0

Delivered

Rate Product

PendingPackedShippedDelivered

Order #2

Name : Swetha Kanumuri

Email : kanumuriswetha9@gmail.com

Address : 903 Avenue C Apt 1302

Total : \$799.0

Ordered On : 2026-07-08 06:35:55.0

Delivered

Rate Product

PendingPackedShippedDelivered

Jewelry World

HomeProducts

My Shopping Cart



Gold Necklace

22K Gold Necklace

\$799.0

Quantity : 1

Subtotal : \$799.0

Remove

Grand Total : \$799.0

Continue ShoppingCheckout

Jewelry World

HomeCollectionsProductsAboutContact



Gold Necklace

5.0/5

\$799.0

Category: Necklace

22K Gold Necklace

Add to CartBuy NowRate Product

Customer Reviews

Swetha Kanumuri

Rating: 5/5

Excellent